

Question	Marking Guidance	Mark	Comments
01.1	1. Cytoplasm 2. Phosphorylation 3. Pyruvate 4. NAD;;	2 max (2 x AO1)	4 correct = 2 marks 2–3 correct = 1 mark 0–1 correct = 0 marks 2. Reject substrate-level or oxidative phosphorylation 3. Accept pyruvic acid 4. Reject NADP but accept any reduced form of NAD e.g. NADH / NADH ₂ / NADH + H

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01.2	Mark in pairs, 1 and 2 OR 3 and 4. (Pair 1 and 2) 1. One carbon dioxide from the link reaction and two carbon dioxide from the Krebs cycle; 2. Link reaction and Krebs cycle occur twice (per glucose) OR Two pyruvate (per glucose); (Pair 3 and 4) 3. (Overall/per glucose) two carbon dioxide from the link reaction; 4. (Overall/per glucose) four carbon dioxide from the Krebs cycle;	2 (2 x AO2)	1, 2, 3 and 4. Accept 'carbon(s)' for carbon dioxide 1. Accept release of carbon dioxide from specified link and Krebs cycle reactions 1, 3 and 4. Accept 'decarboxylation' for 'carbon dioxide' 1 and 2. Accept in a diagram 2. Accept pyruvic acid

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01.3	Mark points 1 and 2 can be awarded in either of the first two sub-headings. (Oxygen uptake decreased)	3 (3 x AO2)	1. Accept less/no oxygen to join to protons/H ⁺ and electrons 2. Ignore 'hydrogen'

	1. (Less/no) oxygen (required as) is final electron acceptor OR (Less/no) oxygen (required as) at end of electron transfer/carrier chain; (ATP production decreased) 2. Less/no oxidative phosphorylation OR Less/no proton/H⁺ movement/gradient OR Less/no chemiosmosis; (Ethanol was produced) 3. Anaerobic respiration occurs OR Pyruvate is reduced (to ethanol);		3. Accept equation showing reduction of pyruvate by reduced NAD to ethanol 3. Ignore fermentation 3. Reject reference to lactic acid
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